



NBA Shot Quality Capstone

Ethical Recommendations & Implications

DAT 490 — Data Science Capstone (2026 Summer A)

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Ethical Recommendations

Our project builds a model that estimates the probability any given shot goes in, then uses that model to measure which players and teams make more shots than expected. The model itself is a measurement tool, and the measurement is sound for what it does. The ethical questions don't come from the model being wrong — they come from what happens when a single number like this gets used to judge people. Shot-quality analysis feeds real decisions in the NBA: who gets signed, who gets traded, how players are paid, and who gets cut. Those are decisions about people's careers and livelihoods, and the responsible use of our work depends on understanding exactly what our metric does and does not capture.

A shot-quality metric measures one thing

The most important thing to understand about our metric is that it only measures shot-making. It asks a narrow, well-defined question — given the context of a shot, did the player make it more often than expected? — and it answers that question well. But basketball value is far wider than shot-making. A center's worth often lives in rim protection, rebounding, and setting screens. A point guard's worth lives substantially in passing, ball-handling, and running an offense. None of that appears anywhere in a shot-quality model, because the model only ever looks at shots that were taken and whether they went in. A player could be one of the most valuable in the league and still rank unremarkably on shot-quality-adjusted scoring, simply because the thing that makes them valuable is not the thing our metric measures.

This is a well-documented danger whenever a single quantitative measure is used to evaluate people. Muller (2018) describes the “tyranny of metrics” as the tendency to replace judgment built from experience with a standardized number, and then to attach real rewards and penalties to that number. The risk is not that the number is meaningless — it is that the number is treated as the whole picture when it only ever captured a slice. Thomas and Uminsky (2020) make the same argument specifically for systems built on machine learning: overemphasizing any single metric leads to a myopic and distorted view, and the appropriate response is to use a slate of metrics together rather than optimizing or evaluating on one in isolation. Our shot-quality metric is exactly the kind of single, well-optimized number their work warns against leaning on too heavily.

What our results actually show

Our analysis makes this concrete. When we rank players by shot-quality-adjusted performance and compare it to traditional field goal percentage, some players move dramatically. Low-usage perimeter players who consistently make difficult, contested shots climb the rankings, while interior players who take and make easy shots near the basket fall. It would be a mistake to read this as the metric being unfair to any group of players. The metric is doing exactly what it was designed to do: it rewards making shots that were hard to make, and it gives little credit for making shots that were easy to begin with. A rim-running center who converts high-probability layups at a solid rate ranks low on shot-quality-adjusted scoring not because the metric is broken, but because finishing easy shots is precisely the thing the metric is built to be unimpressed by.

The ethical point is not that this ranking is wrong — it is that this ranking is narrow. The same center who ranks low on our metric may be enormously valuable for reasons our model never

sees. Treating the shot-quality ranking as a verdict on that player's overall worth would badly misrepresent him, and in a context where these numbers influence contracts and roster spots, that misrepresentation could cost a player real money and opportunities.

Data limitations that affect how the results should be used

Two limitations of our data shape how far the conclusions should be pushed. First, the granular shot-tracking data that makes shot quality measurable — defender distance, shot clock, touch time, and dribbles for each individual shot — is only publicly available for the 2014-15 season. Later seasons restrict that information to aggregated summaries, so our shot-quality model is necessarily built on a single season. Any statement our analysis makes about a specific player is a statement about that player in 2014-15, not a timeless judgment, and it should always be framed that way. Second, the data identifies real, named players and makes public judgments about their shooting. These are public figures and the data is public, which is standard practice in basketball analytics, but it still means our outputs describe real people and should be presented with appropriate care.

Both sides of using the model

None of this means the model shouldn't be used. Used well, shot-quality analysis surfaces genuinely undervalued contributors — the role players who quietly make hard shots that traditional box scores overlook — and it gives teams a fairer way to recognize that kind of contribution. Choosing to ignore shot quality entirely carries its own cost, because teams then keep over-rewarding high-volume scorers who pad their numbers with easy shots and keep underpaying the players who do difficult things well. The ethical line is not whether to use the metric but how. Used as one input among many, it adds real insight. Used as a standalone verdict on a player's worth, it misleads.

Recommendations

Based on this, we recommend the following for anyone using a shot-quality model like ours to evaluate players:

First, shot-quality-adjusted scoring should always be used alongside other measures, never on its own. As Thomas and Uminsky (2020) argue, the right way to use any single metric is as one piece of a broader slate that captures the dimensions a single number cannot. For a basketball player that slate has to include defense, rebounding, passing, and the judgment of people who actually watch the games.

Second, results should always be interpreted in the context of a player's role. A low shot-quality-adjusted score means something completely different for a rim-running center than for a high-usage perimeter creator, and any responsible use of the metric has to account for what a player is actually being asked to do on the floor.

Third, anyone using these numbers in evaluation or negotiation should be transparent with players and their representatives about how they are being measured. The people who build and hold these models have an information advantage over the players whose careers the models affect, and closing that gap is part of using the tool fairly.

Finally, anyone presenting results from a model like this should be honest about its limits: that it is built on a single season of data, that it captures only shot-making and nothing else a player contributes, and that it describes a moment in time rather than a permanent truth. Presented honestly and used as one input among many, shot-quality analysis is a useful addition to how players are understood. Presented as the whole story, it would distort the very thing it is trying to measure.

References

Barocas, S., Hardt, M., & Narayanan, A. (2023). *Fairness and machine learning: Limitations and opportunities*. MIT Press.

Muller, J. Z. (2018). *The tyranny of metrics*. Princeton University Press.

Thomas, R. L., & Uminsky, D. (2020). Reliance on metrics is a fundamental challenge for AI (arXiv:2002.08512) [Preprint]. arXiv.